3. A lift car of weight 8000 N is going up with constant speed 2 m s^{-1} as shown in Figure 3.1. The upward force raising the lift car is provided by the cable wound on a drum which is driven by a motor. The other end of the cable is firmly attached to the drum at P. Neglect air resistance and the mass of the cable.

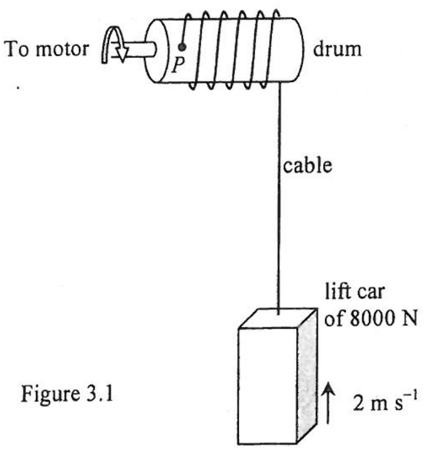

Figure 3.1

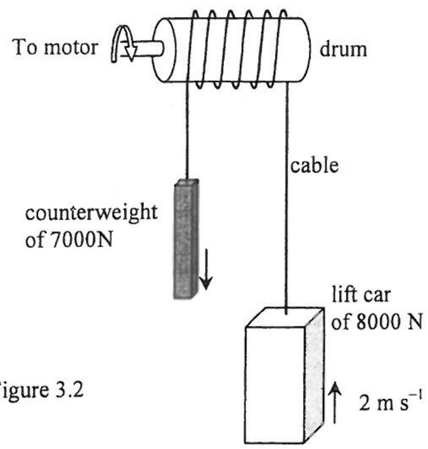

Figure 3.2

(a) (i) Calculate the mechanical power delivered to the rising lift car by the motor. (2 marks)

(ii) The total mechanical power output of the motor is 20 kW. How much power is lost due to overcoming friction between the movable parts ? (1 mark)

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(b) Now a 7000 N counterweight is installed at the other end of the cable as shown in Figure 3.2. The counterweight always moves in the opposite direction to the lift car which again moves up at 2 m s^{-1}. Assume that there is no slipping between the cable and the drum.

(i) Calculate the total mechanical power output of the motor required in this case, assuming the same power loss in overcoming friction between movable parts as found in (a). (2 marks)

(ii) State the advantage of having the counterweight installed. (1 mark)

(iii) A claim is made that as power is lost due to friction, a drum with frictionless surface can further reduce the power required from the motor. Comment on this claim. (2 marks)

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10. (a) In the circuit shown in Figure 10.1, a 12 V battery of negligible internal resistance is connected with a thermistor R and a resistor of resistance 120 Ω. The graph shows the variation of the thermistor's resistance with temperature.

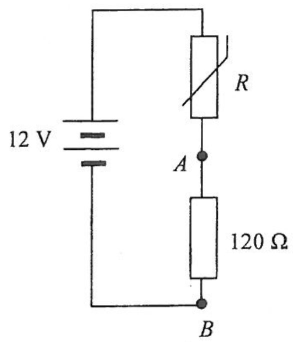
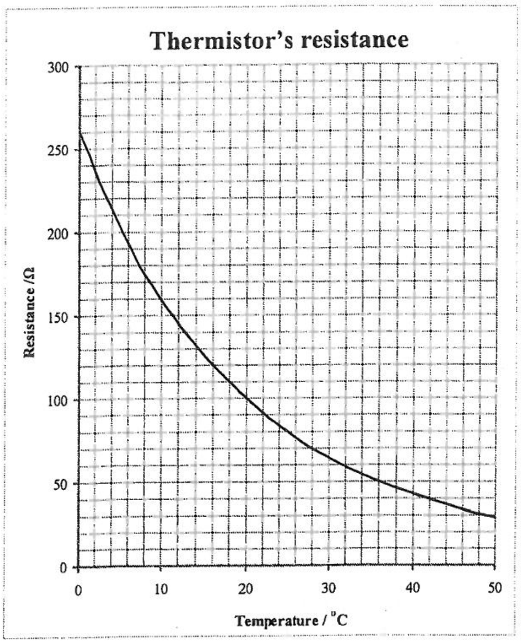


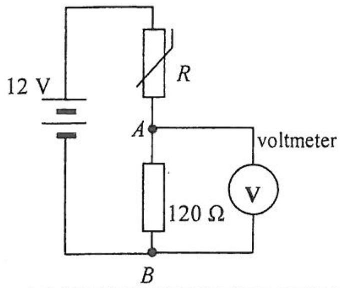
Figure 10.1



(i) Find the resistance of the thermistor R at 25 °C. (1 mark)

(ii) What is the potential difference V_{AB} across A and B at 25 °C ? (2 marks)

(b) Kelly wants to confirm the above calculation by measuring V_{AB} using a voltmeter of about 1 kΩ resistance. She finds that the reading registered is slightly different from the value found in (a) despite making careful measurements. Explain why this is so. Suggest how the accuracy of the measurement could be improved. (3 marks)



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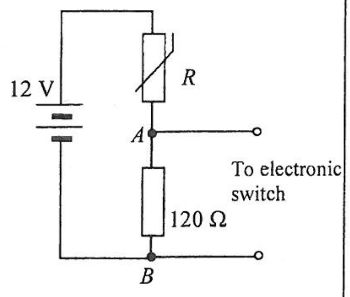
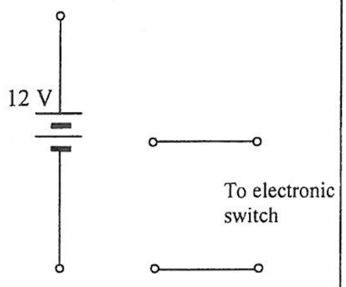
(c) (i) The potential difference V_{AB} is used to drive an electronic switch connected across AB to turn on a fan if temperature rises above a certain value such that V_{AB} is 6.0 V or above. Using the information provided in the graph, find the minimum temperature needed to keep the fan on. Show your working. (2 marks)



(ii) Without using additional components, complete the new circuit diagram below to illustrate how the circuit can be modified to turn on a heating device when temperature falls below a certain value. Explain the action of the circuit. No calculation is required. (3 marks)



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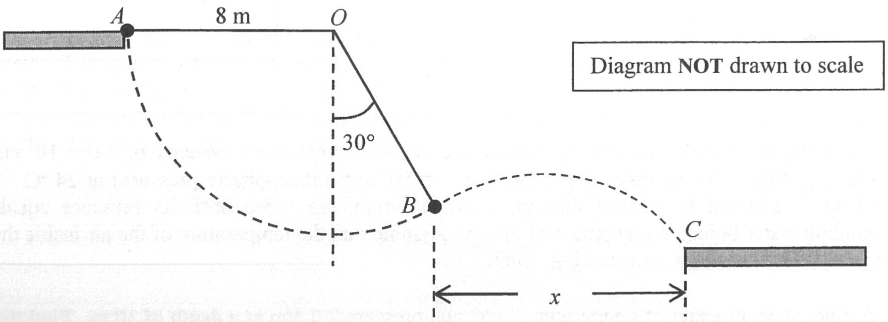

Figure 3.1

Figure 3.1 shows two horizontal platforms with end points A and C. An acrobat tries to swing from A to C by using a light rope of 8 m long and with one end fixed at point O, which is at the same level as A. He leaves A by holding the end of the rope and then releases it when reaching point B at which the angle between the rope and the vertical is 30°. The acrobat can be treated as a point mass and the rope remains taut and not extended throughout the motion. Neglect air resistance. ($g = 9.81\ m\ s^{-2}$)

(a) Mark on Figure 3.1 the velocity v_B of the acrobat at B. If the speed of the acrobat when leaving A is zero, find the magnitude of v_B. (3 marks)

*(b)(i) It takes 1.25 s for the acrobat to reach C after releasing the rope at B. By considering his horizontal motion, find the horizontal separation x between B and C. (2 marks)

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(ii) Calculate the vertical distance of C below B. (3 marks)

(c) Before reaching the lower platform, is there any change to the acrobat's mechanical energy among the points A, B and C ? (1 mark)

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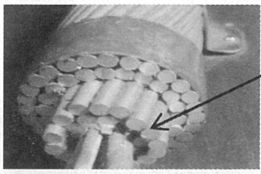
8. Electricity generated from power plants are transmitted at a high voltage through overhead cables in suburban areas.

(a) Each overhead cable consists of 40 strands of identical transmission lines bundled together.



(i) One single strand of transmission line has a cross-sectional area of 1.3×10^{-5} m^2 and resistivity 2.6×10^{-8} Ω m. Find the resistance per km of a single strand of transmission line. (2 marks)

(ii) Explain why the resistance per km of an overhead cable is much smaller than that of a single strand of transmission line. Estimate the resistance per km of an overhead cable. (2 marks)

(iii) Hence, explain why a bird can stand with both feet on a high-voltage cable without getting an electric shock. (2 marks)

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*(b) Electrical power of 180 MW is transmitted at a voltage of 400 kV through an overhead cable.

(i) Calculate the current carried by the overhead cable. (2 marks)

(ii) Show that less than 0.1% of the electrical power is lost after transmitted through a total of 10 km of overhead cable. (2 marks)

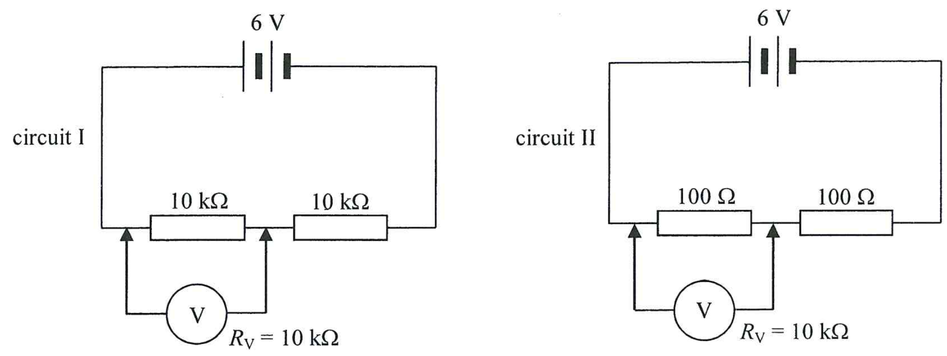
(iii) As the voltage drop across this overhead cable is negligible, a voltage of 400 kV at the cable's end is stepped down by an ideal transformer with turns ratio 12 :1.

(I) Find the secondary voltage from the transformer. (1 mark)

(II) State **ONE** factor leading to energy loss in a practical transformer and suggest the corresponding measure for improvement. (2 marks)

7. (a) The circuits in Figure 7.1 each contains two resistors connected in series with a 6 V battery of negligible internal resistance. The resistors in circuit I are 10 kΩ each while those in circuit II are 100 Ω each.

Figure 7.1



A voltmeter of internal resistance $R_V = 10$ kΩ is used to measure the potential difference across one of the resistors as shown.

(i) What would be the respective voltmeter readings ? (3 marks)

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(ii) In fact, the potential difference across each resistor *before connecting the voltmeter* is 3 V in both circuits. Explain why this voltmeter gives a relatively inaccurate value for circuit I. Hence state the general principle of selecting a suitable voltmeter for such measurement. (2 marks)

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(i) State which reading(s), V_m, I_m or both, do(es) **NOT** give the *true voltage* across the resistor and/or the *true current* passing through the resistor. Hence write down an equation relating R_A, R_m and R. (2 marks)

(ii) Find the percentage error associated with R_m when measuring the resistance of this resistor.
Given: $R_V = 10\ k\Omega$, $R_A = 1\ \Omega$ and $R = 10\ \Omega$. (2 marks)

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4. (a) A steel ball bearing is released from rest at time $t = 0$. A stroboscopic photo is taken at 0.05 s time intervals. The results are shown in Figure 4.1. Neglect air resistance.

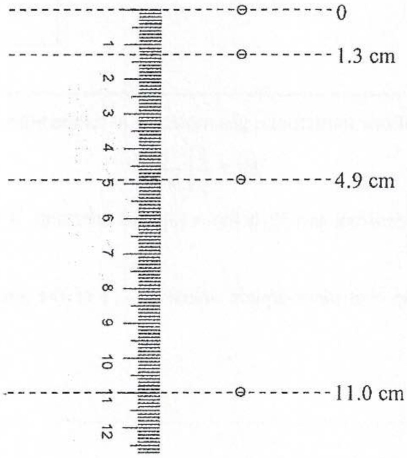

Figure 4.1

(i) Estimate the acceleration due to gravity using the data in Figure 4.1. (2 marks)

*(ii) The bearing is now projected horizontally instead of released from rest. The bearing is projected at time $t = 0$, and a stroboscopic photo is taken at 0.05 s time intervals. The first and the last image of the stroboscopic photo are shown using circles (⊙) in Figure 4.2. For reference, the stroboscopic photo of the bearing released from rest is also shown in the figure using crosses (x).

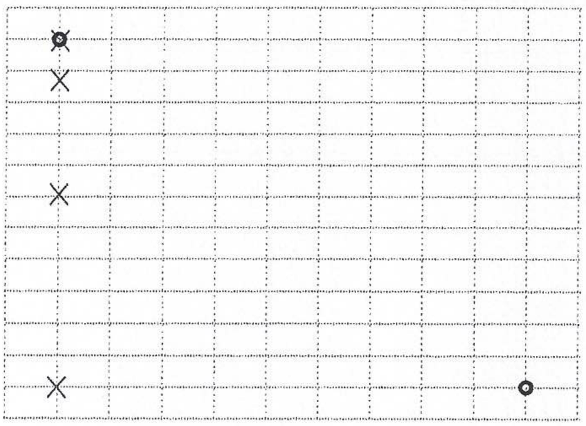
Figure 4.2

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(1) In Figure 4.2, mark the positions of the projected bearing in the stroboscopic photo using circles (⭘). (2 marks)

(2) Given that the bearing is projected horizontally with an initial speed of 1 m s^{-1}, use the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken. (3 marks)

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(b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

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8. A student uses the following apparatus to measure the resistance of a tungsten filament light bulb.

a battery, a switch, a variable resistor, an ammeter, a voltmeter, a light bulb

(a) Figure 8.1 shows an incomplete circuit for the experiment. The '+' symbol represents the positive terminal of the ammeter.
Use suitable circuit symbols to complete the circuit, and mark the positive terminal of the voltmeter with '+'. (3 marks)

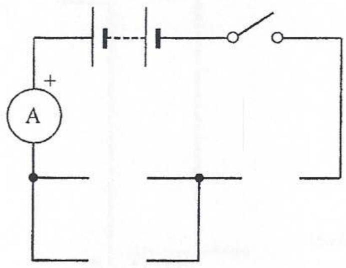

Figure 8.1

The table below and Figure 8.2 show the results obtained.

Voltage across the light bulb V / V	0	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0
Current I / mA	0	76	112	126	133	139	170	226	273

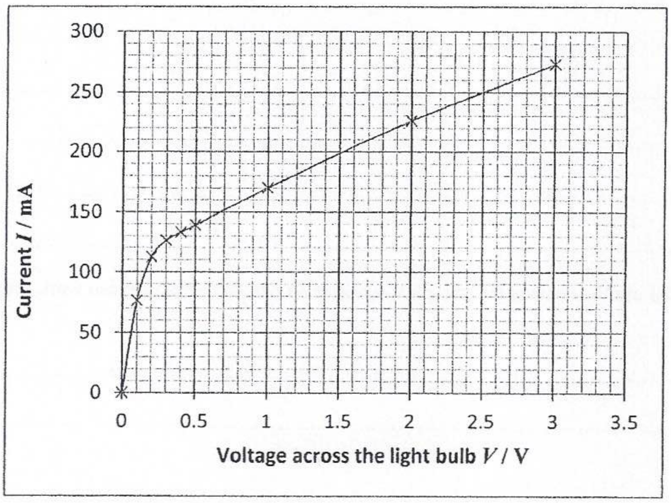

Figure 8.2

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(b) Briefly explain the variation of the resistance of the light bulb with the voltage across the light bulb. (2 marks)

(c) The student claims that since the resistance of the light bulb is not a constant, the equation $R = V/I$ cannot be used to calculate the resistance of the light bulb. Briefly explain why his claim is wrong. (1 mark)

(d) Determine the resistance of the light bulb at $V = 0.1$ V and 2.5 V. (3 marks)

(e) It is given that the cross-sectional area of the tungsten filament in the light bulb is 1.66×10^{-9} m^2, and the resistivity of tungsten at room temperature is about 5.6×10^{-8} Ω m. Estimate the length of the tungsten filament in the light bulb using the appropriate resistance found in (d). (3 marks)

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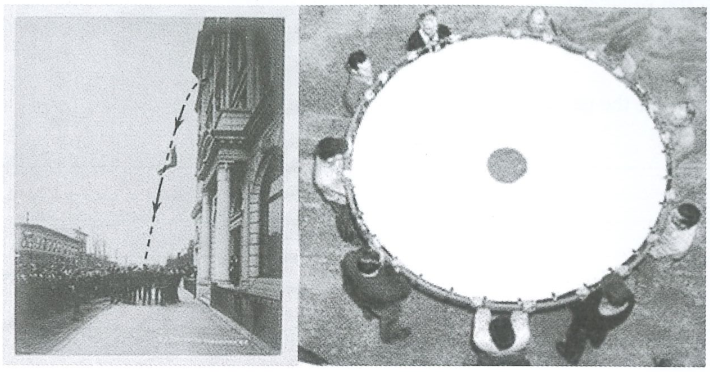
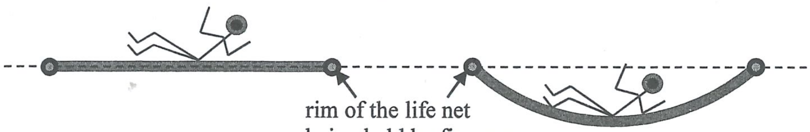
3. Read the following passage about **life nets** and answer the questions that follow.

A life net is a rescue equipment formerly used by firefighters. It gives people on the upper floors of a burning building an opportunity to jump to safety, usually to ground level. It became obsolete due to advances in firefighting technology.



The practical height limit for successful use of life nets is about six storeys, although a few people once have survived jumps from an eight-storey building into a life net with various degree of injuries. The diagrams below explain its working principle.



When a person hits the net, it deforms and puts the person to a stop in a longer time as compared to hitting the solid ground.

(a) A person falls from a height of 12 m above a life net with negligible initial speed. Neglect air resistance and the size of the person. ($g = 9.81\ \text{m s}^{-2}$)

(i) Estimate (1) the vertical speed v and (2) the time of fall t of the person just before hitting the life net. (4 marks)

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(ii) If this falling person of mass 70 kg is stopped in 0.3 s by the life net, estimate the average force acting on the person by the net within this time interval. (3 marks)

(iii) What form of energy is stored by the life net during the deceleration of the falling person ? (1 mark)

(b) (i) Give a reason why there exists a height limit of using life nets. (1 mark)

*(ii) The falling person might hit the rim of the net, thus the person or the firemen holding the rim would be injured. Explain why it is not easy for a person jumping from a height to reach the life net's central part. (2 marks)

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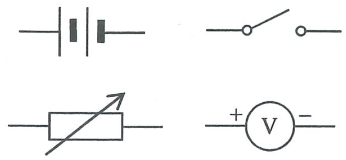
7. You are provided with a battery (of fixed e.m.f. ξ and internal resistance r), a variable resistor (with several known resistance values R to be selected), a switch, a voltmeter (assumed ideal) and a few connecting wires.



(a) With the aid of a circuit diagram, describe the procedure of an experiment to study how the terminal voltage V delivered by the battery depends on the resistance R connected to it. State **ONE** precaution of the experiment. (5 marks)

(b) Describe the variation of V with R and express V in terms of ξ, r and R. (2 marks)

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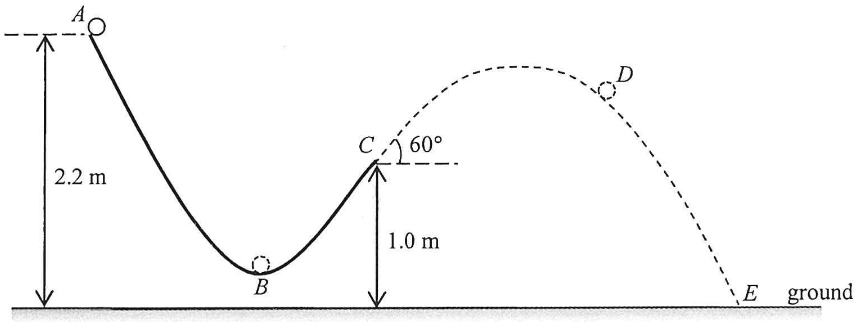
4. A small sphere is released from rest at point A and runs along a smooth track ABC as shown in Figure 4.1. The track around the lowest point B is approximately circular in shape.

Figure 4.1



The sphere leaves the track at point C where the track makes an angle of 60° with the horizontal. It finally reaches point E on the ground. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

(a) Arrange the speeds of the sphere at points A, B, C and D respectively in descending order. (1 mark)

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*(b) On Figure 4.1, use arrows to indicate the acceleration of the sphere, if any, at point B and at point D respectively. (2 marks)

(c) (i) Describe the energy conversion of the sphere when it goes along the track ABC. (2 marks)

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(ii) Hence find the speed of the sphere at point C. (2 marks)

*(iii) If the horizontal distance between points C and E is 2.55 m, calculate the time of flight of the sphere before reaching point E. (3 marks)

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4. A trolley is connected to a hanger of mass 20 g by a light inextensible string as shown in Figure 4.1. Four slotted weights, each of mass 20 g, are loaded onto the trolley. The experiment is designed to investigate the relationship between the net force acting on the system (trolley and slotted weights with hanger) and its acceleration. The acceleration a is measured after the trolley is released on the smooth horizontal runway.

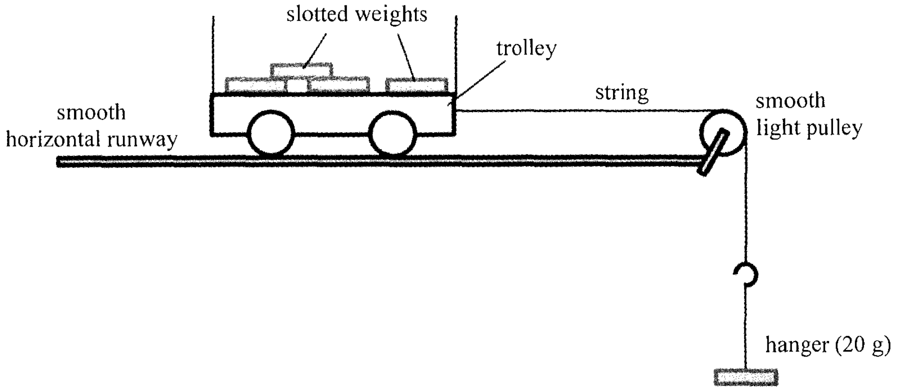
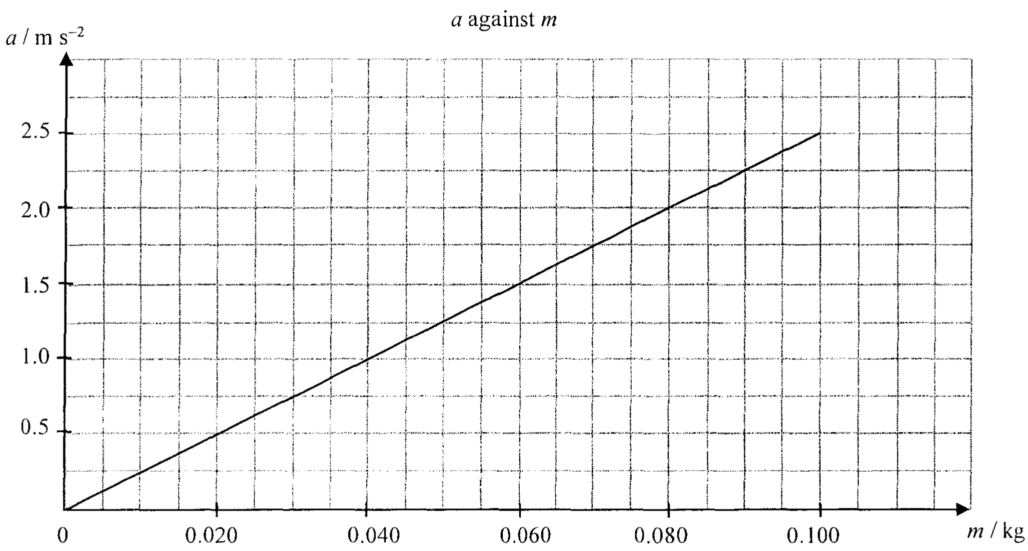


Figure 4.1

The experiment is repeated by transferring the slotted weights one by one from the trolley to the hanger so as to increase the mass hanging, m.

no. of weights transferred to the hanger	0	1	2	3	4
mass hanging m / kg	0.020	0.040	0.060	0.080	0.100

The results obtained are used for plotting a graph of a against m as shown below. Neglect both air resistance and the frictional forces acting on the trolley. ($g = 9.81$ m s^{-2})



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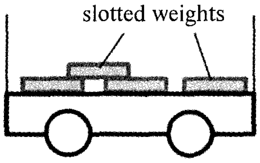
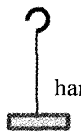
(a) (i) After the trolley is released, indicate in the figures below (1) the horizontal force(s) acting on the loaded trolley, and (2) the force(s) acting on the hanger. (2 marks)





(ii) Is the tension in the string equal to, greater than or smaller than the weight of the mass hanging when the system is released ? Explain. (2 marks)

(iii) By considering the motion of the whole system, or otherwise, write an equation relating m, a and mass M of the trolley. (1 mark)

(b) Calculate the slope of the graph. Hence find M using the result of (a)(iii). (3 marks)

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5. A rocket carrying an artificial satellite is launched vertically from the Earth. When the rocket is at a certain height from the Earth's surface, it expels 2.60×10^3 kg of gas per second with a certain speed v towards the Earth's centre. As a result, an average thrust of 5.20×10^6 N is produced. Neglect air resistance.

(a) (i) Assuming that the speed of the rocket is negligible, estimate v. (2 marks)

(ii) At a certain instant, the total mass of the rocket and the artificial satellite is 3.60×10^5 kg while the acceleration due to gravity at the rocket's position is 8.56 m s^{-2}. Estimate the acceleration a of the rocket at this position. (2 marks)

(iii) Suppose the rocket keeps expelling gas at the same rate for a few seconds. Would the rocket's acceleration increase, decrease or remain unchanged in that duration ? Explain. (2 marks)

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*(b) The artificial satellite is put in the geostationary orbit of radius r around the Earth. It appears to be always stationary above an observer at the equator.

(i) State the period of this satellite. (1 mark)

(ii) Show that r is approximately 42000 km. ($g = 9.81\ \mathrm{m\ s^{-2}}$) (2 marks)
Given: radius of the Earth = 6.37×10^{6} m

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8.

Figure 8.1

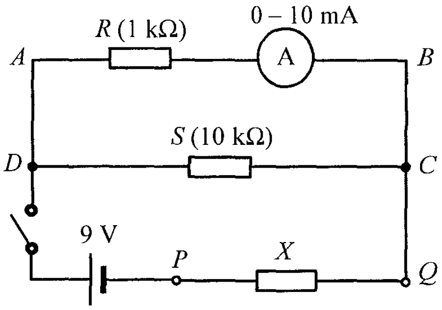

Figure 8.1 shows a circuit for measuring the resistance of resistor X connected across P and Q. The resistance of resistor S is 10 kΩ. The internal resistance of the 9 V cell and that of the ammeter are negligible.

(a) When the switch is closed, the ammeter reads 8.5 mA.

(i) What is the p.d. between A and B ? (2 marks)

(ii) Find the current passing through resistor S. (2 marks)

(iii) Indicate on Figure 8.1 the direction of current in each of the three branches via C. (2 marks)

(iv) Deduce the p.d. across resistor X. Hence, find the resistance of X. (3 marks)

(b) State the purpose of connecting resistor R in series with the ammeter. (1 mark)

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Figure 4.1

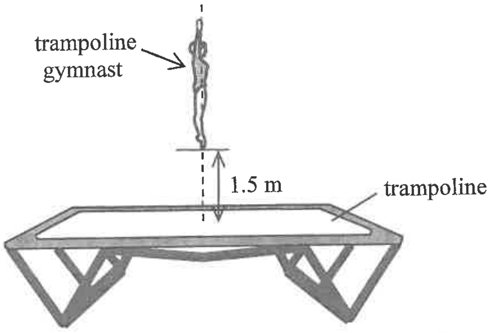

Figure 4.1 shows a trampoline gymnast of mass 50 kg performing a straight jump. Her feet are 1.5 m above the trampoline at the maximum height. Neglect air resistance and assume that the gymnast maintains this posture throughout the jump. ($g = 9.81\ \text{m s}^{-2}$)

(a) Find the kinetic energy of the gymnast just as her feet touch the trampoline on the way down from her jump. (2 marks)

(b) After touching the trampoline, the gymnast keeps on moving downward for 0.40 m further before she stops.

(i) Describe the energy transfer to the trampoline by the gymnast **after touching the trampoline.** (2 marks)

(ii) Estimate the average force exerted by the gymnast on the trampoline. (2 marks)

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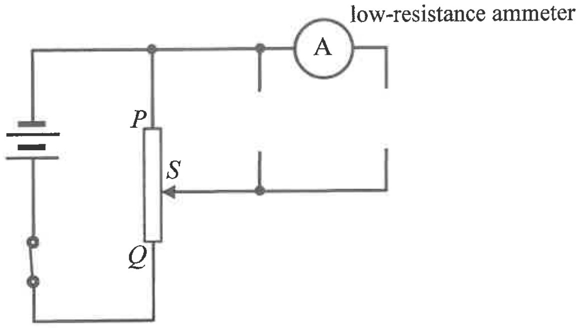
8. (a) A student sets up the circuit in Figure 8.1 to find the current-voltage (I-V) characteristic of a filament light bulb.

Figure 8.1



PQ is a variable resistor in the circuit with sliding contact S. The bulb and a high-resistance voltmeter are missing in the circuit.

(i) Use suitable circuit symbols to complete the circuit. (1 mark)

(ii) How would the brightness of the light bulb change when contact S is adjusted from P to Q ? (1 mark)

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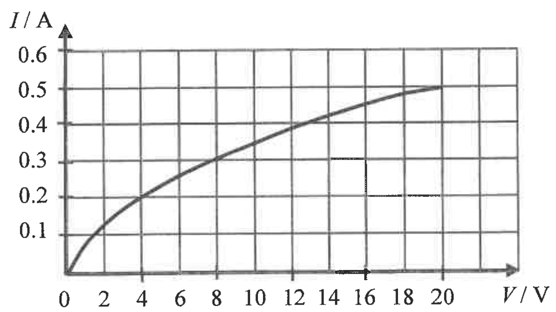
(b) The graph below represents the I-V characteristic of this light bulb which has a rated voltage of 20 V.

Figure 8.2



(i) Find the resistance of the bulb when working under its rated voltage. (2 marks)

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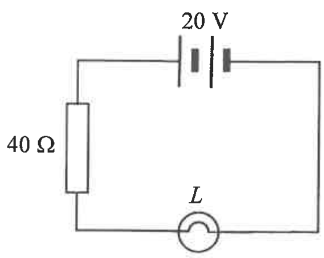
(ii) Explain why the resistance of the bulb varies with the applied voltage V. (2 marks)

(c) The bulb L in (b) and a 40 Ω resistor are connected in series with a 20 V battery of negligible internal resistance as shown in Figure 8.3.

Figure 8.3



The current I in the circuit and the voltage V across the bulb is related by $I = 0.5 - 0.025V$.

(i) Add a straight line on Figure 8.2 to determine the current I. (2 marks)

(ii) Hence estimate the power consumed by bulb L. (2 marks)

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7. (a) A straight metallic wire of length 0.20 m and cross-sectional area 8.0×10^{-7} m^2 has a resistance of 0.50 Ω. Find the resistivity, in Ω m, of the material that the wire is made of. (2 marks)

(b) Four such metallic wires in (a) are joined together to form a square coil *CDEF*. The coil is connected to a circuit consisting of a 1.5 V cell of negligible internal resistance and two identical resistors R_1 and R_2, each of 2.0 Ω, as shown in Figure 7.1.

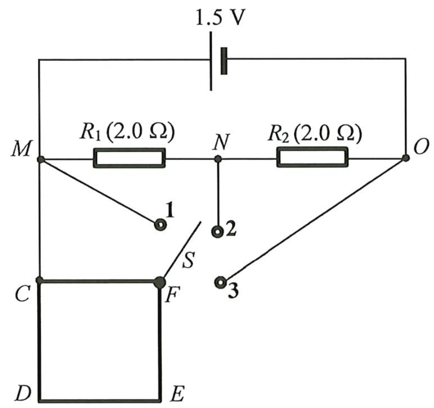

Figure 7.1

(i) To which terminal (**1**, **2** or **3**) should switch *S* be connected in order to have a maximum current flowing through side *CF* of the coil ? (1 mark)

(ii) With *S* connected to terminal **2**, find the equivalent resistance across *MN*. (2 marks)

Answers written in the margins will not be marked.

(iii) With S connected to terminal **2**, is the potential difference across R_2 greater than, smaller than or equal to that across R_1 ? (1 mark)

(iv) With S connected to terminal **1**, what is the power dissipated by the coil ? Give your reason. (2 marks)

0

11. Read the following passage about **deformation sensors** and answer the questions that follow.

Figure 11.1(a) shows a simple sensing device consisting of a thin, flat rectangular metal foil attached to a plastic backing plate. When this foil, length l and width w, is stretched to deform as shown in Figure 11.1(b), its resistance will change.

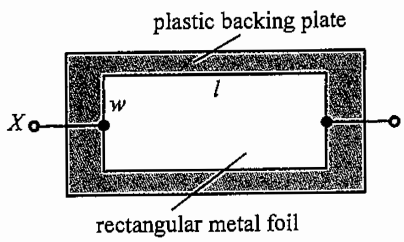

Figure 11.1(a)

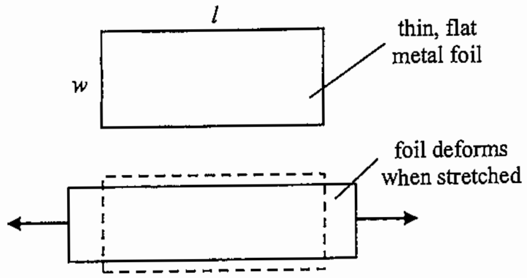

Figure 11.1(b)

When this device is connected to a suitable circuit, it can serve as a sensor to monitor deformation which results in a change in parameters of the circuit (such as voltage).

(a) Referring to Figure 11.1(a) and (b), state and explain how the metal foil's resistance across XY changes when the foil is stretched. Assume that the thickness and resistivity of the foil remain unchanged. (2 marks)

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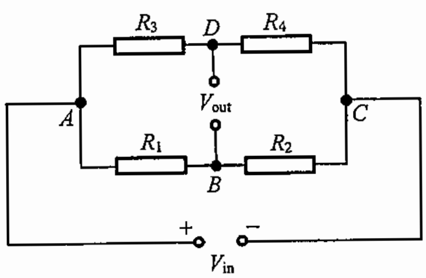
(b) Figure 11.2 shows a circuit with four resistors of resistance R_1, R_2, R_3 and R_4 connected to an input voltage V_{in}. V_{out} is the voltage across DB.

Figure 11.2



(i) Current I_1 flows through branch ABC. Write an equation relating V_{in}, I_1, R_1 and R_2. (1 mark)

(ii) R_4 in the circuit is the sensing device described in the passage, with unstretched resistance 500 Ω. Find the corresponding percentage change in V_{out}/V_{in} when R_4 increases 0.2% (i.e. becomes 501 Ω).

Given: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$; $R_1 = R_2 = R_3 = 470\ \Omega$. (2 marks)

(iii) Sensing devices S_1, S_2, S_3 and S_4 connected to a suitable circuit are attached to ceiling $ABCD$ as shown in Figure 11.3 in order to monitor the growth of the cracks ①, ② and ③. After a few weeks, the extent of deformation detected is in the order $S_1 > S_2 > S_3 > S_4$.

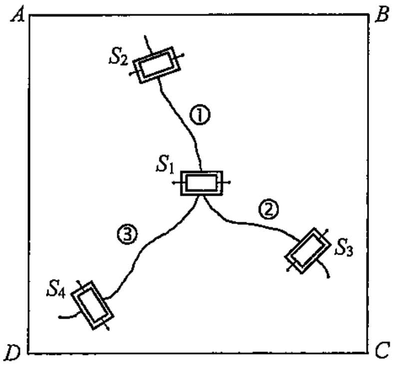


Figure 11.3

(I) Deduce which crack (①, ② or ③) has grown the most. (1 mark)

(II) Deduce the most probable stretching direction (AB, BC or AC) along the ceiling. (1 mark)

Answers written in the margins will not be marked.

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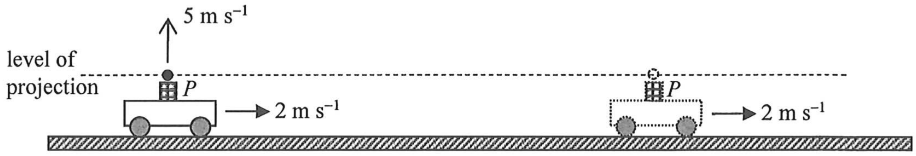
2. A trolley is moving on a horizontal surface with a constant velocity of 2 m s^{-1}. At time $t = 0$ s, a small bead of mass 0.01 kg is projected upwards with a vertical speed of 5 m s^{-1} from a device P mounted on the trolley as shown. Neglect air resistance.

Figure 2.1



*(a) Find the time taken, t, for the bead to travel back to the level of projection. (2 marks)

(b) What is the kinetic energy of the bead at its maximum height ? (2 marks)

*(c) When the bead falls back to the level of projection, it is just above device P as shown. Explain why the bead does not fall behind or beyond device P at that moment. (2 marks)

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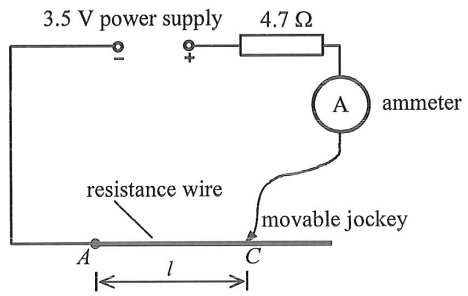
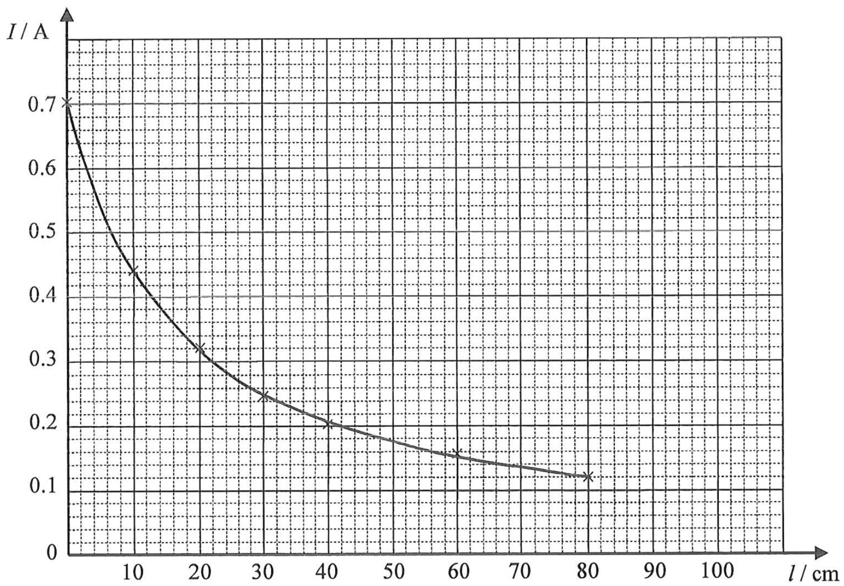
Answers written in the margins will not be marked.

7. The circuit shown in Figure 7.1 can be used to measure the current at different lengths of a resistance wire using a movable jockey. The power supply of negligible internal resistance delivers a voltage of 3.5 V.

Figure 7.1



When the jockey is placed at point C on the resistance wire, the length of wire used is l and the corresponding current I is measured with an ammeter. The experiment is repeated at different lengths l and a graph of I against l is plotted below.



(a) What is the function of the 4.7 Ω resistor in this experiment ? (1 mark)

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(b) When the jockey is placed at point A (i.e. $l = 0$), find the resistance of the whole circuit. Compare it with the value of the 4.7 Ω resistor and account for the difference. (3 marks)

(c) When the jockey is moved from point A on the resistance wire to point C where $l = 30$ cm, find the increase in resistance of the circuit. (2 marks)

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11. A coal-fired power plant consumes 4.5×10^6 kg of coal per day to generate electric power at an average rate of 680 MW. Given that burning 1 kg coal releases 32 MJ of energy.

(a) (i) Find the daily energy input (in MJ).

(2 marks)

(ii) Find the electrical energy produced per day (in MJ). (2 marks)

(iii) Estimate the efficiency of this power plant. (2 marks)

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(b) Transmission cables at 400 kV are used for long-distance electric power transmission to a city.

(i) Estimate the current in the cables. (2 marks)

(ii) Estimate the power loss (in kW) in the transmission cables. The resistance of the cables is $2.90 \times 10^{-8}\ \Omega$ per metre and the total length of the cables is 160 km. (2 marks)

*(iii) Explain why a.c. is preferred for electric power transmission. (2 marks)

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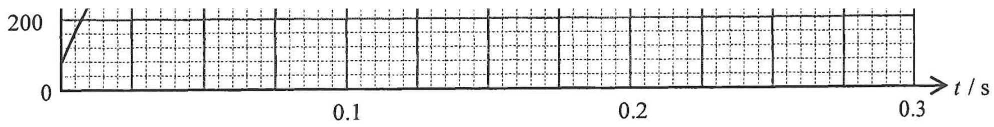

(a) The bulb works at its rated value after R becomes stable. Estimate the rated power of the bulb. (2 marks)

(b) Explain why R initially increases and becomes stable after a period of time. (3 marks)

(c) The filament of the bulb has a length of 0.56 m and a uniform cross-section of diameter 0.022 mm. Find the resistivity of the filament at $t = 0$ s. (2 marks)

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Answers written in the margins will not be marked.

11. (a) A spacecraft with an astronaut on board is launched on a rocket. The rocket with the spacecraft has an initial mass of 4.80×10^5 kg at take-off. The rocket engine expels hot exhaust gas at a constant speed of 2600 m s^{-1} downwards relative to the rocket. Assume that 1.15×10^3 kg of gas is expelled in the first 0.5 s. (Neglect air resistance.)

(i) Calculate the average thrust (the upward force) acting on the rocket due to the exhaust gas during the first 0.5 s. (2 marks)

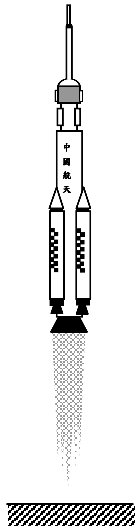

Figure 11.1

(ii) On Figure 11.1, draw and label an arrow for each force acting on the rocket. Assuming that the change in mass of the rocket during the first 0.5 s is negligible, estimate the acceleration of the rocket. (3 marks)

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(b) The spacecraft of mass 7.80×10^3 kg now enters a circular orbit of radius r around the Earth.

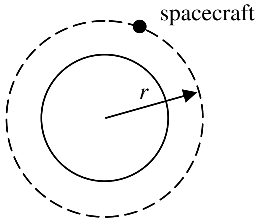

Figure 11.2

*(i) Show that the speed of the spacecraft in the orbit is given by $\sqrt{\frac{g}{r}} R_E$ where R_E is the radius of the Earth. (2 marks)

*(ii) How long does it take for the spacecraft to orbit the Earth 14 times ? (3 marks)
Given : radius of the orbit $r = 6.71 \times 10^6$ m
radius of the Earth $R_E = 6.37 \times 10^6$ m

(c) Give **ONE** reason why an aircraft is unable to fly in space like a rocket. (1 mark)

Answers written in the margins will not be marked.